

Power Reduction in Continuous-Time Delta-Sigma Modulators Using the Assisted Opamp Technique

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Abstract—The opamp in the first integrator of a high resolution single-bit continuous-time $\Delta\Sigma$ modulator has stringent slew rate requirements, which increases power dissipation. We introduce the “assisted opamp” integrator, which is a way of achieving low distortion operation with low power consumption. We present circuit implementations of our technique for single-bit modulators using NRZ and switched-capacitor-resistor (SCR) feedback DACs. Audio modulators designed in a $0.18\ \mu\text{m}$ CMOS technology are used as vehicles to demonstrate the effectiveness of our techniques. The modulator with an NRZ DAC achieves a dynamic range of 92.5 dB in a 24 kHz bandwidth and dissipates $110\ \mu\text{W}$ from a 1.8 V supply. A second design, which employs an SCR-DAC, achieves a dynamic range of 91.5 dB and dissipates $122\ \mu\text{W}$. The figures of merit (FOM) of these modulators, 175.9 dB and 174.4 dB respectively, are comparable with those of state-of-the-art multibit designs.

Index Terms—Analog-to-digital converter, CMOS process, continuous-time circuit, continuous-time integrator, digital audio, digital-to-analog converter, distortion reduction, linearity enhancement, loop filter, oversampling, sigma delta, switched capacitor, 0.18 micron.

I. INTRODUCTION

CONTINUOUS-TIME delta-sigma modulation (CTDSM) is attractive due to several features like implicit antialiasing, relaxed speed requirements on the active elements and resistive input impedance (which eases the design of the stage driving the ADC). Several modulators targeting the audio range have been reported recently, where power reduction is the main motivation for choosing CTDSMs over their discrete-time counterparts [1]–[6]. Early designs (the first three of the references cited above) used a single-bit quantizer. Subsequent designs have largely employed multibit quantizers. When compared to single-bit CTDSMs, multibit designs are less sensitive to clock jitter and relax the linearity requirements of the loopfilter. However, a single-bit design has several attractive features. The quantizer is simple, resulting in reduced area/power. Dynamic element matching (DEM) circuitry is not needed, further reducing area and complexity.

To leverage the implementation advantages of a single-bit design, one needs to address the issues of clock jitter sensitivity and loop filter linearity requirements. It is well known that jitter affects the performance of CTDSMs by perturbing

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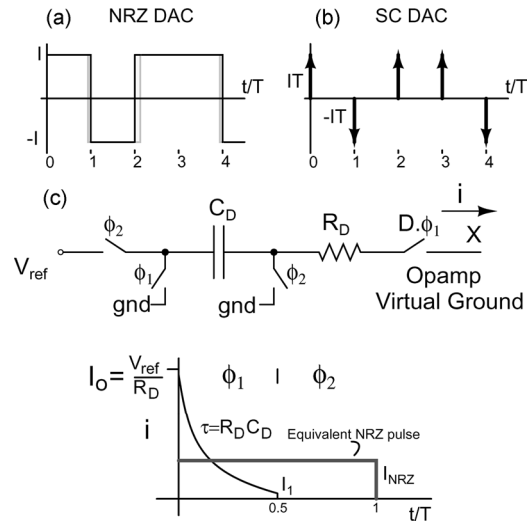


Fig. 1. (a) NRZ DAC feedback waveform, (b) SC DAC feedback waveform, (c) SCR DAC schematic, and the exponentially decaying pulse produced by the DAC.

the width of the feedback DAC pulse, making an NRZ DAC preferable to an RZ DAC. If the modulator output sequence and the jitter in the n^{th} clock edge are denoted by $y[n]$ and $\Delta T[n]$ respectively, an NRZ DAC injects an error proportional to $\Delta T[n](y[n] - y[n-1])$, as shown in Fig. 1(a). One way of reducing noise due to clock jitter is through appropriate PLL design. The full scale jumps in the feedback DAC signals also demand excellent linearity of the first integrator in the loop filter (thereby increasing power dissipation). Another way of addressing clock jitter is to inject a fixed charge into the loop filter using an impulsive DAC (implemented using a switched-capacitor (SC), Fig. 1(b)). It can be shown that noise due to clock jitter is first-order noise shaped, and is therefore negligible in the signal band. If the NRZ DAC is problematic for the linearity of the first integrator, an SC-DAC is even more so, since the first integrator has to now source/sink impulsive currents. It is perhaps for this reason that SC-DACs have not found much favor with CTDSM designers. A compromise is to use an exponentially decaying DAC pulse [2], generated as shown in Fig. 1(c). In ϕ_1 , the capacitor C_D (which has been previously charged to V_{ref} during ϕ_2) discharges into the virtual ground of the opamp through a resistor R_D with a time-constant $\tau = C_D R_D$. The initial (peak) current injected by the DAC is $I_0 = V_{\text{ref}}/R_D$, decaying to $I_1 = I_0 \exp(-T/2\tau)$. If $I_1 \ll I_0$, the average current is given by $C_D V_{\text{ref}} f_s$ (which is the height of the equivalent NRZ DAC pulse), and $I_1 < C_D V_{\text{ref}} f_s$. This type of DAC pulse generator has also been referred to as the SCR DAC in the literature. The noise introduced by this DAC due to clock jitter

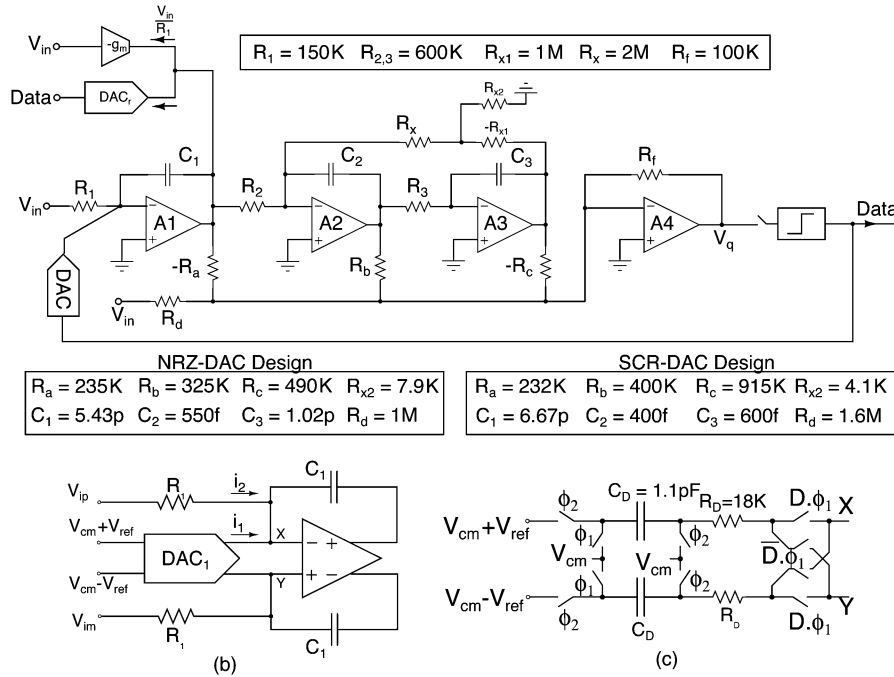


Fig. 2. (a) Modulator architecture and component values. (b) Schematic of the first integrator in the modulator with the SCR DAC. (c) Schematic of the differential SCR DAC.

is proportional to I_1 [2], and is controlled by T/τ . Choosing a small τ (small R_D) reduces jitter sensitivity, but results in an increased peak current, which in turn increases the linearity requirements of the first integrator. The choice of R_D , therefore, involves a tradeoff between linearity and jitter noise.

From the discussion above, it is seen that designing the first integrator in a single-bit modulator with adequate linearity and low power is a challenge, especially with an SCR DAC. In this work, we introduce a circuit technique called the *assisted opamp integrator* that addresses this problem [7]. Thanks to this approach, we are able to achieve 15 bit performance in single bit CTDSMs based on NRZ and SCR feedback DACs with power dissipation that is comparable to that of the multibit modulator we reported in [6]. Implemented in the same $0.18\ \mu m$ CMOS technology used in [6], the NRZ/SCR modulators consume 110/122 μW from a 1.8 V supply and achieve dynamic ranges of 92.5/91.5 dB for a signal bandwidth of 24 kHz. The modulators operate with an OSR of 128. The architectural and circuit details of these modulators form the subject of the rest of the paper, which is organized as follows. In Section II, we justify the various architectural design choices made in this work. The assisted opamp technique is developed in Section III. Section IV gives circuit design details. Experimental results from prototype modulators, designed in a $0.18\ \mu m$ CMOS process, are shown in Section V. Section VI concludes the paper.

II. MODULATOR ARCHITECTURE

A single-ended equivalent of the modulators, and the respective component values are shown in Fig. 2(a). In the figure, negative resistors are used to indicate the inversion of the signals driving them. The targeted signal bandwidth and sampling rate are 24 kHz and 6.144 MHz, respectively. The NTF is chosen

to be maximally flat, with an out-of-band gain (OBG) of 1.5 and zeroes spread in the signal band. The full scale voltage of the modulator is 3.6 V (peak-to-peak differential). The resulting in-band peak SQNR is about 110 dB, which is well above the desired 92 dB SNR of the converter. The performance of the designs are therefore limited by thermal noise. The loop filter is implemented as a “Cascade of Integrators with Feed Forward” (CIFF) structure. For low noise and high linearity, active-RC techniques are used to implement the integrators. The first integrator uses 150 K Ω resistors, while 600 K Ω resistors are used in the second and third integrators.

Some aspects of the architecture are the following.

- *Direct Path to the Quantizer:* A direct path is added from the modulator input to the loop filter output through R_d . This technique, originally proposed to reduce distortion in discrete-time DSMs [8] is adopted in our continuous-time design. The benefit of using the direct path is the reduction in the values of the integrating capacitors (especially C_3), since the signal component of the loop filter output is being provided by the modulator input.

- It is well known that a CIFF implementation of the loop filter results in a signal transfer function (STF) with peaking, as well as poorer rejection in the alias bands $[(mf_s - f_b) - (mf_s + f_b)]$, when compared to a modulator design with a CIBF loop filter. A potential issue with the addition of the direct path is the effect on the antialiasing property of the modulator—and one might be tempted to infer that this attribute of the modulator is lost. This is not so, as can be seen from the following. If the modulator input is a tone at a frequency $f_s + \Delta f$, the output of the loop filter largely comprises of the contribution from the direct path (since the integrator unity gain bandwidths are much smaller than f_s). This tone, injected through the direct path, is sampled and undergoes noise shaping, just like quantization

noise. Thus, the tone appears at the modulator output as a frequency Δf (due to aliasing), with an amplitude $g|NTF(\Delta f)|$, where g denotes the gain of the direct path. Since $|NTF(\Delta f)|$ is very small, and the gain of the direct path is of the order of unity, the alias rejection is good.

- It is interesting to determine the loss in the alias rejection incurred with respect to a CIFF modulator without the direct path. In the discussion that follows, $\hat{L}_c$ and L_c denote the transfer functions from the DSM input to the quantizer input ($V_q(s)/V_{in}(s)$ in Fig. 2(a)), for modulators with and without the direct path respectively. From Fig. 2(a), we see that L_c and $\hat{L}_c$ around f_s (the sampling rate) can be approximated by

$$L_c(j2\pi f) \approx \frac{1}{j2\pi f C_1 R_1} \frac{R_f}{R_a}, \quad (\text{no direct path})$$

$$\hat{L}_c(j2\pi f) \approx \frac{1}{j2\pi f C_1 R_1} \frac{R_f}{R_a} + \frac{R_f}{R_d}, \quad (\text{with direct path})$$

From the above equations, we see that the direct path increases the gain of the filter at high frequencies, affecting the alias rejection. Thus, the degradation in the alias rejection around f_s is given by

$$\left| \frac{\hat{L}_c(j2\pi f_s)}{L_c(j2\pi f_s)} \right| \approx \frac{2\pi f_s C_1 R_1 R_a}{R_d} \quad (1)$$

For the component values chosen in Fig. 2, the STF magnitude (around f_s) is increased by 17.4 dB and 14.9 dB in the NRZ-DAC and SCR-DAC designs, respectively. In spite of this degradation, the alias-rejection is still quite good (about -85 dB at f_s)—thereby justifying the use of the direct path to reduce capacitor area.

- *Implementation of the Notch in the NTF*: Realizing NTF complex zeros (which improve the SQNR by about 8 dB) is traditionally done using a single feedback resistor around two integrators (in the notation of Fig. 2, $R_{x1} = 0$ and $R_{x2} = \infty$). For a third-order NTF and 24 kHz bandwidth, calculations show that using a single resistor to realize the notch necessitates an impractically large R_x (of approximately 200 M Ω !). To avoid this, the notch is implemented using a T-network formed by R_{x1} , R_{x2} and R_x . The notch frequency is given by $\omega_z^2 \approx (R_{x2}/R_{x1})(1/R_3 R_x C_2 C_3)$. R_{x1} and R_{x2} are realized without any overhead, as they are reused as the common-mode detecting resistors needed anyway at the output of A3. In this design, the in-band quantization noise is about 18 dB lower than thermal noise—thus variations in the notch frequency have a negligible impact on the in-band SNR.

- *Separate Summing Amplifier versus Capacitive Summation in the Third Integrator*: A CIFF loop filter has several advantages with respect to suppressing noise and distortion from sources other than the first integrator (at the high OSR used in this work). However, feed-forward requires a summing operation to be performed. Two well-known techniques to accomplish addition of the integrator outputs are: (a) capacitive feed-ins into the third integrator, or (b) use of a dedicated summing amplifier. In the capacitive summation approach, the first integrator needs to drive a capacitive load; as explained in the following section, this increases modulator distortion. Mitigating this necessitates increased power dissipation in A1. Further, the third integrator is now in the high speed path (the $1/s$ path of the loop filter)

of the modulator, and the feed-in capacitors needed to implement summation result in increased delay when compared to the delay of a stand-alone integrator. Hence, even though capacitive summation needs only three opamps, it is not clear that it saves power. This was the reasoning behind using a separate summing amplifier (opamp A4).

- Finally, thanks to the single-bit quantizer, the output of the summing amplifier can be scaled without changing the NTF—this can be used to advantage in the design of A4, since it does not need to support full rail swings.

- *The Assisted Opamp Integrator*: The first integrator is implemented in an unconventional manner (see the replica DAC (DAC_r) and transconductor injecting current into the output of the first opamp). As shown in the next section, this approach significantly reduces distortion in the first integrator.

- *DACs*: The NRZ-DAC based design uses a resistive DAC for low excess noise. The schematic of the first integrator of the SCR-DAC based modulator is shown in Fig. 2(b). The details of the DAC are shown in part (c) of the figure. The capacitors C_D are charged to $\pm V_{ref}$ in ϕ_2 . In ϕ_1 , they are discharged into the virtual grounds (X and Y) depending on the quantizer output D . The discharge time-constant is given by $(R_D + 2R_s)C_D$, where R_s denotes the resistance of the two switches in series with C_D . As discussed in the introduction, the choice of R_D is a tradeoff between jitter sensitivity and opamp slew-rate requirements. In this work, $C_D = 1.1$ pF and $R_D + 2R_s = 22.5$ k Ω , resulting in a time constant of about 25 ns. The DAC pulse shape is shown in Fig. 1(c)—the peak current is 40 μ A, decaying to about 1.5 μ A in half clock cycle (which is ≈ 3.3 time constants of the SCR DAC). The average current is 6 μ A.

The first integrator uses an opamp with a pMOS input stage for low $1/f$ noise. The other integrators and the summing amplifier use operational amplifiers with nMOS input pairs. The input-referred noise of the loop filter is dominated by the thermal noise from the DAC and the first integrator. The opamps are two-stage designs that use feedforward compensation. In this design, the influence of the finite speed of the opamps on the NTF is negligible. The excess delay in the quantizer is about 100 ps. Since the NTF has a relatively low out of band gain of 1.5, this delay is of little consequence. RC time constants in the loop filter vary due to process shifts and changes in ambient temperature. To counter this, the integrating resistors and capacitors are implemented as digitally tuned banks. These banks, which are externally controlled, can compensate for RC shifts of up to $\pm 25\%$ from the nominal, and bring them to within $\pm 5\%$.

III. THE ASSISTED OPAMP INTEGRATOR

The discussion in Section I indicates that the design of the first integrator with adequate linearity and low power dissipation represents a significant challenge in the design of high performance single bit CTDSMs. In this section, we introduce a technique that addresses this problem. To motivate our solution, consider the first integrator in a conventional single-bit modulator, as shown in Fig. 3(a). A single-ended circuit, assuming an NRZ feedback DAC is shown for simplicity—the same arguments hold for an SCR-DAC. The input is a sinusoid, with an amplitude V_{max} . The feedback DAC waveform takes on one of two values $\pm V_{ref}$. If the opamp is ideal, the current i_1 flowing

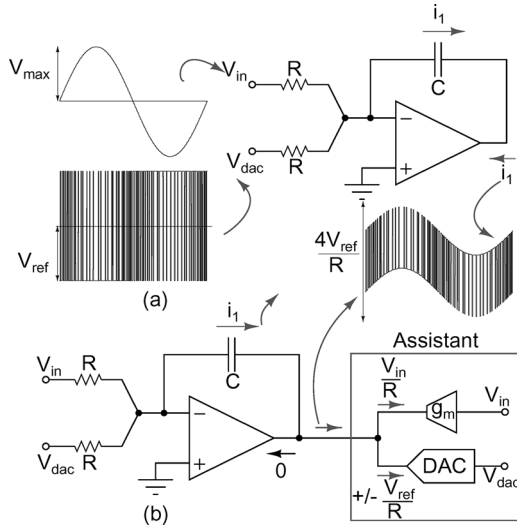


Fig. 3. (a) Schematic and waveforms in a conventional integrator and (b) the assisted opamp integrator.

through the integrating capacitor is as shown in the figure. The peak-to-peak value of this current is almost $4(V_{\text{ref}}/R)$. The current is seen to mostly consist of very high-frequency content, as one would expect due to the voltage steps introduced by the feedback DAC. The virtual ground voltage remains at zero, and the in-band component of V_{dac} equals $-V_{\text{in}}$.

A real opamp, however, cannot respond fast enough to the sharp steps of the DAC feedback voltage. This effect manifests as jumps in the virtual ground potential whenever the feedback DAC switches. If the opamp is linear (even though it has finite bandwidth), these jumps do not result in distortion. With a practical opamp, the large jumps at the virtual ground node result in nonlinear behavior and degrade the performance of the modulator. The conventional way of addressing this issue is to bias the opamps with large currents, so that the bandwidth and slew rate of the opamp are enhanced. It is thus seen that the inability of the opamp to source or sink large currents instantaneously is responsible for the nonlinearity in the modulator. This suggests the following solution. The current demanded of the opamp is known exactly, since one has access to V_{in} and V_{dac} . This information can be used to “assist” the opamp in discharging its function, as shown in Fig. 3(b). A transconductor with $g_m = 1/R$ and a replica current steering DAC with outputs $\pm(V_{\text{ref}}/R)$ (referred to as the assistant) pull current out of the opamp—this

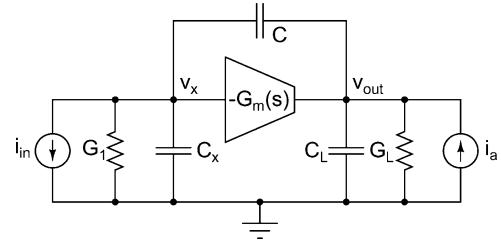


Fig. 4. Equivalent circuit of an assisted opamp integrator— i_{in} is the input current, and i_a is the assistant.

way, the opamp does not have to supply any current. Since the demands made on the opamp are now greatly relaxed, the excursions of the virtual ground voltage are reduced, resulting in low distortion operation. The steps of current are no longer a problem, since they are supplied by the assistant.

A more detailed diagram of the assisted opamp integrator, considering practical nonidealities is shown in Fig. 4. i_{in} and G_1 denote the Norton equivalent of the input and DAC circuits. C_x is the capacitance at the virtual ground v_x . C is the integrating capacitor, while C_L and G_L are the total load capacitance and conductance that the opamp needs to drive. The opamp itself is modelled as a frequency-dependent transconductor $G_m(s)$. This is an accurate representation of single-stage and two-stage feedforward compensated opamps. i_a denotes the assistant current. The distortion introduced by the integrator is dependent on the swing at the virtual ground. The Laplace transform of v_x is shown in (2) at the bottom of the page. From the equation, we see v_x will not swing if

$$I_a(s) = \left(1 + \frac{C_L}{C}\right) I_{\text{in}}(s) + I_{\text{in}} \frac{G_L}{sC}. \quad (3)$$

Zero swing at v_x means ideal integrator behavior with zero distortion. From (3), we see that the assistant current comprises of proportional and integral versions of the input current. In a sigma-delta modulator, the current through G_L is usually small (in comparison with the current flowing through the integrating capacitor) since subsequent integrators are impedance scaled to reduce power dissipation. If subsequent integrators and the summation of their outputs are implemented using $G_m - C$ techniques, $G_L = 0$. Then, (2) and (3) can be simplified as (4) and (5), shown at the bottom of the page. Note that the cancellation at v_x does not depend on C_x , G_1 or the shape of i_{in} . Fig. 5(a) and (b) shows the implementation of the “assisted opamp” idea for integrators based on single-stage

$$V_x(s) = \frac{-I_{\text{in}}(s)(s(C + C_L) + G_L) + I_a(s)sC}{G_1 G_L + sC(G_m(s) + G_L(1 + \frac{C_x}{C}) + G_1(1 + \frac{C_x}{C})) + s^2(C_x C + C_x C_L) + C C_L} \quad (2)$$

$$V_x(s) \approx \frac{-I_{\text{in}}(s)s(C + C_L) + I_a(s)sC}{sC(G_m(s) + G_1(1 + \frac{C_x}{C})) + s^2(C_x C + C_x C_L + C C_L)} \quad (4)$$

$$I_a(s) \approx \left(1 + \frac{C_L}{C}\right) I_{\text{in}}(s) \quad (5)$$

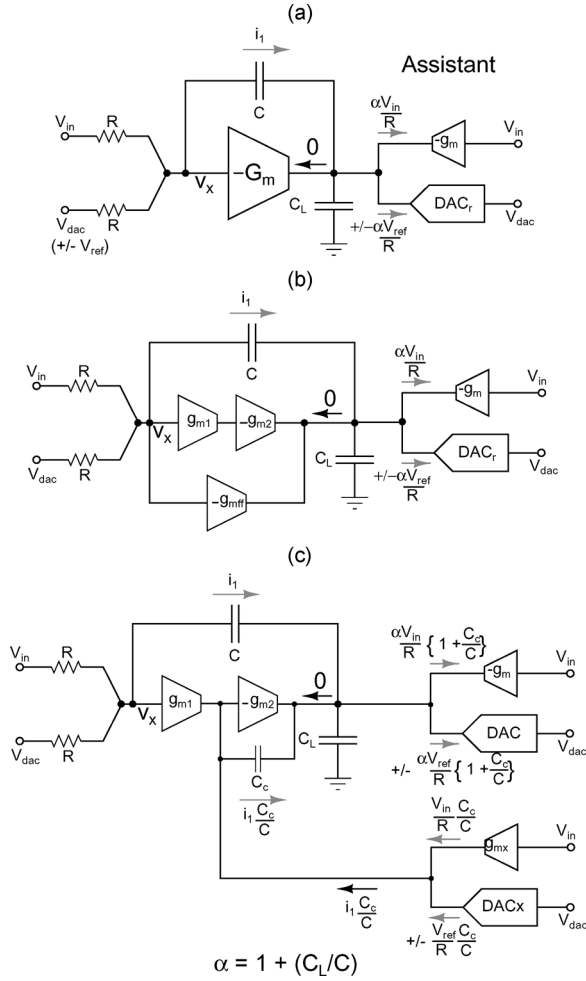


Fig. 5. Implementation of the assisted opamp integrator for several commonly used opamp topologies: (a) a single-stage OTA; (b) a two-stage feedforward compensated OTA; (c) a Miller-compensated OTA.

and two-stage feedforward compensated opamps. The assistant current i_a is realized using a transconductor (that provides the input component of the integrator current) and a current steering DAC (which provides the DAC component of the integrator current). A Miller-compensated two-stage opamp can also be “assisted”, as shown in Fig. 5(c). Two assistants are needed in this case—apart from the one that injects current into the output, another assistant is needed to remove the current $i_i(C_c/C)$ that flows through the compensating capacitor. This way, g_{m1} does not need to sink any current, resulting in $v_x = 0$.

From the discussion above, it is seen that opamp assistance results in a dramatic reduction in the swing at the virtual ground node of the opamp, thereby significantly improving the distortion performance of the integrator (and in consequence, the modulator). Since the opamps do not source/sink any current (thanks to the assistants), they can be designed with lower quiescent currents, thereby saving power.

• **Noise due to Opamp Assistance:** In a CTDSM, the in-band input-referred noise of the opamp degrades the performance of the modulator. Apart from supplying the desired signal current, the assistant circuitry injects thermal and $1/f$ noise. The spectral density of the noise current injected by the assistant is about $8 kT/R$ (assuming the noise contributed by each arm

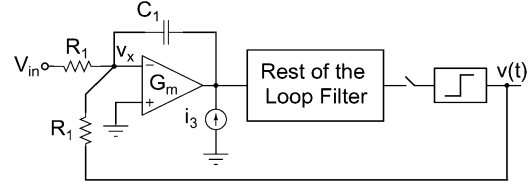


Fig. 6. Simplified diagram of the modulator used to evaluate the effect of distortion of the assistant transconductor.

of the assistant corresponds to a transconductance of $1/R$.) In the single-stage OTA case with assistance [Fig. 5(a)], this extra noise source appears in parallel with the noise of the OTA. In a well-designed integrator, $G_m \gg 1/R$; therefore, it is seen that noise injected by the assistant degrades the input-referred opamp noise only slightly.

• In the two-stage opamp using feedforward compensation [Fig. 5(b)], the noise current due to the assistant appears in parallel with that of the second stage of the opamp. Thus, the noise due to the assistant (and g_{m2}) become negligible when referred to the opamp input.

• For a Miller-compensated two-stage opamp [Fig. 5(c)], the noise injected by the assistant circuitry appears in parallel with that injected by g_{m1} and g_{m2} . The source occurring at the output of g_{m1} could be of potential concern. It turns out that this will not be an issue in a well-designed integrator, where the gain-bandwidth product of the opamp is chosen to be much higher than the unity gain bandwidth of the integrator. This results in

$$g_{m1} \gg \frac{1}{R} \frac{C_c}{C}. \quad (6)$$

The right-hand side of the equation above can be recognized to be g_{mx} shown in Fig. 5(c), indicating that the noise penalty due to the assistant is negligible. The noise injected by the assistant at the opamp output is of no consequence, like in the feedforward example.

• **Distortion of the Assistant Circuitry:** In practice, the assistant transconductor will be weakly nonlinear and its current can be modelled as $i_{a1} = (\alpha/R)V_{in} - \alpha g_3 V_{in}^3$, where $V_{in}/R \gg g_3 V_{in}^3$. Denoting the input amplitude by A and the harmonic distortion of the assistant by $HD_{3,a}$, we see that harmonic current tone injected by the assistant has an amplitude given by

$$I_{3a} = HD_{3,a} \frac{\alpha A}{R} = \frac{\alpha g_3 A^3}{4}. \quad (7)$$

The analysis given below holds for integrators that use single-stage and two-stage feedforward compensated opamps, and a CIFF modulator. The modulator input (and its third harmonic) are low-frequency signals, and we assume that the rest of the modulator does not load the first integrator. The effect of distortion of the assistant on the output spectrum can be understood using the simplified diagram shown in Fig. 6, where i_3 denotes the harmonic current injected by the assistant. Very little (≈ 0) current flows through C_1 at low frequencies (thanks to the high unity-gain bandwidth of the first integrator), so all of i_3 must be “absorbed” by the opamp. This must result in a harmonic component in the virtual ground voltage v_x , with an amplitude

$$A_{vx,3} = \frac{I_{3a}}{G_m} = \frac{\alpha g_3 A^3}{4 G_m} \quad (8)$$

where G_m is the low-frequency transconductance of the opamp. Applying KCL at v_x at the third harmonic frequency (and noting that the current through C_1 at this frequency is nearly zero), we see that $v(t)$ must have a component at the third harmonic with an amplitude given by

$$A_{v,3} = \frac{\alpha g_3 A^3}{2G_m}. \quad (9)$$

The third harmonic distortion in the modulator output is

$$\text{HD}_3 = \frac{\alpha g_3 A^2}{2G_m} = \frac{2\alpha \text{HD}_{3,a}}{G_m R}. \quad (10)$$

It is thus seen that the distortion of the assistant transconductor is reduced by $G_m R/2$ (which is large by design—for example, in the feedforward compensated opamp, $G_m = g_{m1} r_{o1} g_{m2}$). Simulations show that transconductor $\text{HD}_{3,a}$ as large as 0.5% results in virtually distortion free modulator performance. We exploit this by using a class-AB design for the transconductor. The current steering DAC, whose distortion is also attenuated by $G_m R/2$, is easy to design for low distortion operation.

• **Mismatch and Timing Errors in the Assistant Circuitry:** A deviation in the magnitude of the assistant current from the desired value of $i_{in}(1 + (C_L/C))$, and timing skew due to finite bandwidth effects will result in imperfect cancellation, and as seen from (2) the swing at v_x will not be zero. Intuitively, it is seen that mismatch/skew can be tolerated, as long as the swing at v_x does not become large enough to push the transconductors into their strongly nonlinear regions of operation. Quantitative estimation of mismatch and skew effects appears difficult and depends on details of the opamp design. A guideline for design, which is apparent from (2), is to make $G_m(s)$ large. Simulation results given in the next section show that mismatch and skew are not problematic in our designs.

Based on the discussion in this section, a two-stage feedforward topology was chosen to implement the opamps. Such a topology is more efficient when compared to a Miller-compensated design, since power is not “wasted” by charging and discharging the compensating capacitor [9]. Further, assisting a Miller opamp is more complicated, as seen from Fig. 5(c). A single-stage opamp is not preferred since the G_m realizable for a given bias current is small. Finally, opamp assistance is only used in the first integrator since distortion introduced by subsequent circuitry is attenuated by the high in-band gain of the first integrator. This is an inherent advantage of the CIFF loop filter architecture, and the high OSR ($= 128$) being used.

IV. CIRCUIT DESIGN AND SIMULATION RESULTS

A. Operational Amplifiers and Quantizer

The two-stage feedforward compensated opamp used in the first integrator of the NRZ-DAC based modulator is shown in Fig. 7. The first stage consists of long channel devices M_1, M_2, M_3 and M_4 to lower input-referred $1/f$ noise. M_5 and M_6 provide common-mode feedback for the first stage. These are minimum length devices, carrying 20% of the first stage tail current. Since $M_{9,10}$ and $M_{5,6}$ have the same V_{DS} , the quiescent current in the second stage is accurately set to $4 \mu\text{A}$ by choosing M_9 to be 8 times wider than M_5 . The reason for the choice of $4 \mu\text{A}$ is the following. The modulator coefficients are scaled in

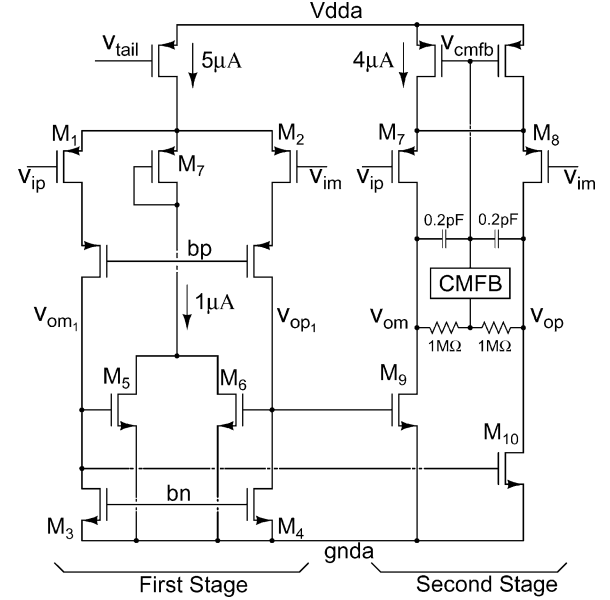


Fig. 7. Operational amplifier used in the first integrator.

such a manner that $1.5 \mu\text{A}$ is needed from the first opamp to drive the resistive load current for subsequent stages. $2.5 \mu\text{A}$ is allocated for design margin. The total current drawn by the opamp is $15 \mu\text{A}$. Note that the second stage current is reused to provide the feedforward transconductor. The simulated DC gain and unity gain bandwidth of the opamp (when loaded by $150 \text{ K}\Omega$ resistors) are 65 dB and 55 MHz, respectively.

A similar opamp was used in [9], where resistive common-mode feedback was used at the output of the first stage. A resistive common-mode detector degrades the DC gain and increases area occupied by the opamp. It also constrains the current density in the first and second stages to be the same. Our design uses transistors to detect the common-mode voltage, thereby maintaining the inherently high DC gain of the first stage. The current density of $M_{3,4}$ can be set independently of that in the output stage—allowing them to be sized based on $1/f$ noise considerations, while $M_{5,6}$ and $M_{9,10}$ are optimized for speed. The opamp used in the SCR-DAC based modulator is similar to that in Fig. 7—the only difference is that the quiescent current in the second stage is increased to $6 \mu\text{A}$ per leg. The reason for this is the mismatch expected between the assistant and feedback DAC pulse shapes, and will become clearer in the next subsection. The opamps used in the subsequent integrators and the summing block are identical, and use feedforward compensation. Since $1/f$ noise is not an issue in these opamps, nMOS input and feedforward pairs are used, while g_{m2} is implemented using pMOS devices. The current consumed per opamp is $8.5 \mu\text{A}$.

The circuit diagram of the comparator and the corresponding clock waveforms are shown in Fig. 8(a) and (b), respectively. The comparator operates as follows. When LC is high, the back-to-back inverters formed by M_1, M_2, M_3 , and M_4 are disabled, and the differential input is sampled on nodes X and Y. After LC falls, the regeneration phase begins when L goes high. The decision of the latch is held by two C2MOS (clocked CMOS) inverters which sample the latch output after a small delay.

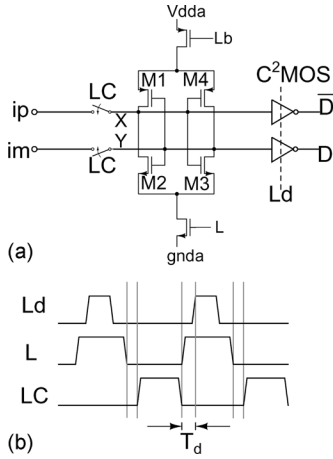


Fig. 8. (a) Comparator schematic and (b) clock waveforms.

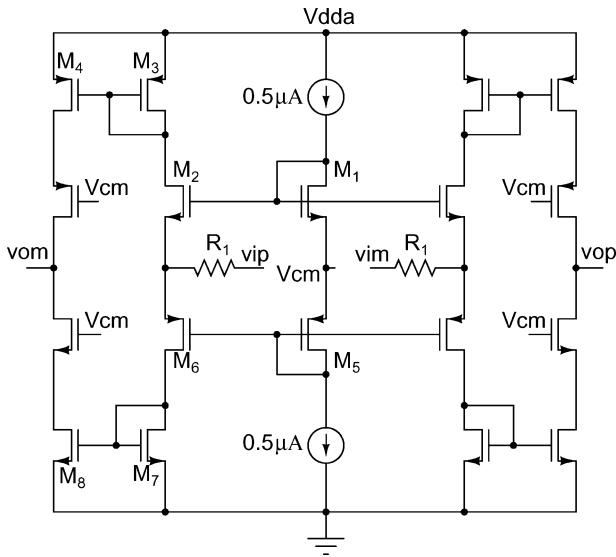


Fig. 9. Transconductor used in the assistant circuit.

B. Assistant Circuitry

Assistant Transconductor: Fig. 9 shows the schematic of the assistant transconductor used in the first integrator. It is a class-AB design, comprising of complementary common gate stages M_2 and M_6 which carry $0.5 \mu\text{A}$ of current under no signal conditions. R_1 is chosen to be 300 K . M_4 and M_8 are chosen to be twice as large as M_3 and M_7 , respectively. As discussed in the previous section, distortion components generated by this transconductor are attenuated by the first integrator, and are of little consequence.

Assistant DAC in the NRZ Design: The assistant DAC consists of $6 \mu\text{A}$ cascoded nMOS and pMOS current sources steered using differential pairs. The current sources are derived from a current proportional to V_{ref}/R , where R denotes the resistance of the modulator feedback DAC. Thus, the magnitude of the assistant DAC current tracks variations of the feedback DAC current over process and temperature.

Assistant DAC in the SCR Design: As discussed in the introductory section, the exponentially decaying SCR-DAC feedback pulse has a peak current of about $40 \mu\text{A}$. The assistant

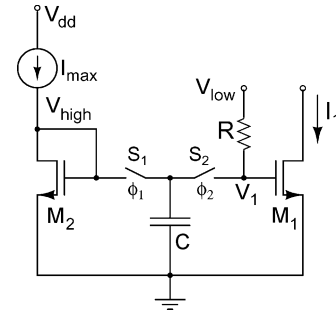


Fig. 10. Simplified schematic of an exponentially decaying pulse generator proposed in [10].

SCR-DAC needs to produce a pulse-shape that matches that of the feedback DAC robustly over process and temperature, and represents a significant circuit design challenge.

A possible choice for the assistant DAC is the circuit proposed by the authors of [10]. Their circuit is shown in Fig. 10, and operates as follows. For simplicity, M_1 and M_2 are assumed to be identical devices. During ϕ_1 , the capacitor C is charged to V_{high} . In ϕ_2 , it is discharged to $V_{\text{low}} = V_{TN}$ (the threshold voltage of M_1/M_2) through a resistor R . The choice of V_{TN} for V_{low} is to avoid having to charge C all the way from zero during ϕ_1 . If the switches are ideal, the gate voltage of M_1 (denoted by V_1) is an exponentially decaying pulse with a time constant $\tau = RC$. The drain current of M_1 , which is initially I_{max} , decays to zero as V_1 approaches V_{TN} . If the transistors are assumed to behave like square-law devices, it is easy to show that the time constant of the decaying current is 0.5τ . Several problems were encountered while attempting to adopt this technique in our design, as described below.

- **Error in initial current:** When ϕ_2 goes high, V_1 should initially be V_{high} so that $I_1 = I_{\text{max}}$. Several mechanisms prevent this from happening. Charge sharing between C and the gate capacitance of M_1 causes a reduction in V_1 , which is aggravated by the voltage drop across switch S_2 . Threshold mismatch between M_1 and M_2 leads to an error proportional to the g_m of M_1 (since the initial current is high, the g_m and current error are also large).

- **Error in decay time constant:** Since M_1 is not a true square law device, the decay time constant of I_1 does not track the time constant of the feedback DAC pulse.

- **Error in the off state:** If V_{low} is larger than V_{TN} (due to mismatch), I_1 does not go to zero.

From the discussion in the previous section, it was seen that good matching between the initial current of the SC feedback DAC and the assistant are necessary to prevent excessive swing at the virtual ground of the opamp. Unfortunately, using the circuit of Fig. 10 as the assistant DAC results in significant error in the initial portions of the feedback pulse. While the issues described above are problematic in our design, they are not of much consequence in [10], as is apparent from that reference.

The fundamental reason for the inaccuracy of the circuit of Fig. 10 is that all error mechanisms directly impact the V_{GS} of M_1 . The exponentially decaying pulse generator used in our work desensitizes the output pulse with respect to the properties of the transistor by switching the capacitor at the source

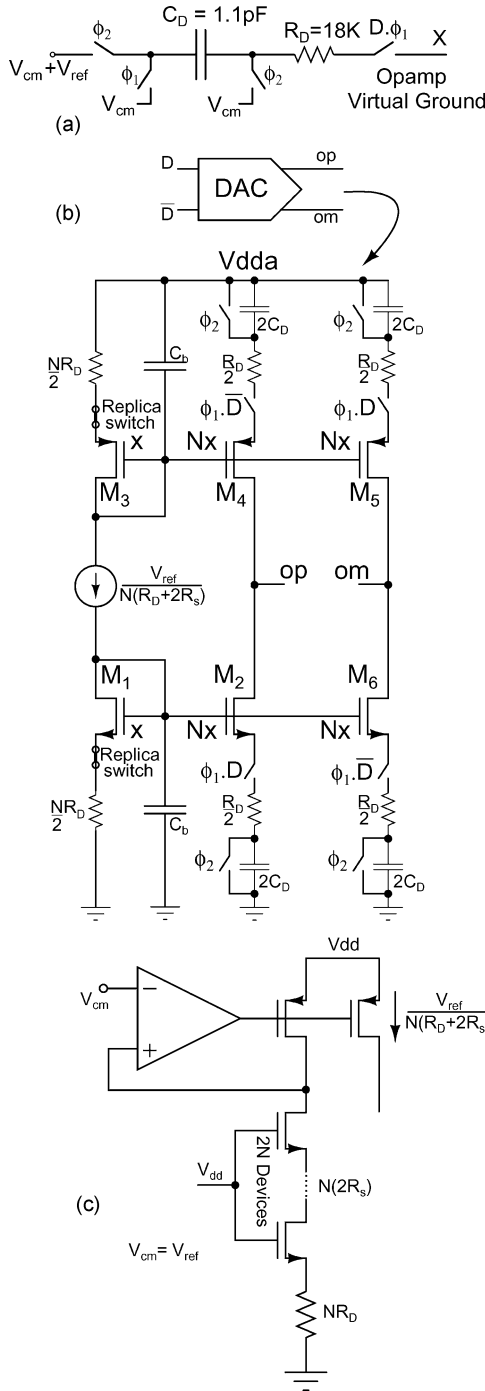


Fig. 11. (a) Single branch of the SCR feedback DAC. (b) Schematic of the SCR assistant DAC. (c) Simplified schematic of the bias current generator.

of the device (akin to the benefits of source degeneration). The functioning of the proposed assistant circuit is explained with the help of Fig. 11. Part (a) of the figure shows one branch of the SCR-feedback DAC, which consists of a capacitor $C_D (= 1.1 \text{ pF})$ that discharges into the opamp virtual ground through $R_D (= 18 \text{ k}\Omega)$ and the on resistance of two switches, denoted by $2R_s (= 4.5 \text{ k}\Omega)$. Fig. 11(b) shows the assistant DAC. The operation of the nMOS portion is described below—the pMOS section operates in a similar fashion. M_2 , M_6 are $N (= 40)$ times wider than M_1 , which is biased with $I_{\text{ref}} = V_{\text{ref}} / (R_D + 2R_s)$. During ϕ_2 , the capacitors denoted by $2C_D$ are discharged.

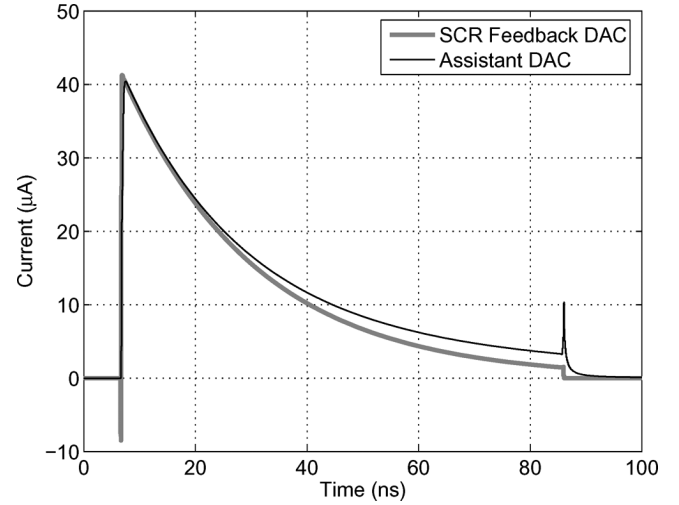


Fig. 12. Pulse shapes of the SCR-feedback and assistant DACs.

In ϕ_1 , the switches at the sources of either M_2 or M_6 are turned on, depending on the quantizer decision D . The drain current of M_2 or M_6 (which should not go into the triode region during modulator operation) starts out at $V_{\text{ref}} / (R_D + 2R_s)$, and decays exponentially toward zero with a time constant $(R_D + 2R_s)C_D$ as the capacitor $2C_D$ keeps charging. Replica switches are used in series with the sources of M_1 and M_3 to match the peak current/time constant of the assistant and feedback DACs. The differential component of the current pulses injected by the feedback and assistant DACs are shown in Fig. 12. Notice the good matching between the two pulses when the injected current is large. At the end of ϕ_2 , the assistant pulse deviates from the feedback pulse by about $1.8 \mu\text{A}$ due to the decreasing g_m of M_2/M_6 . This is not a problem as the difference is small—the current in the second stage of the opamp is deliberately increased to $6 \mu\text{A}$ per leg to be able to handle this mismatch. Capacitors C_b (approximately 5 pF) are needed for high-frequency bypass, and are realized using MOS gates. The average current drawn by the assistant DAC from the supply can be shown to be $C_D V_{\text{ref}} f_s (= 6 \mu\text{A})$. Our design is less sensitive to charge injection and device/layout parasitics when compared to prior art, where the gate of a grounded source device is excited with an exponentially decaying pulse.

The reader might wonder why $2C_D$ and $0.5R_D$ are used to set the decay time constant in the assistant DAC, when the use of C_D and R_D could result in a smaller area. The reason for our choice is the following. Note that M_2/M_6 and M_4/M_5 should not operate in the triode region. Using C_D and R_D would result in a higher(lower) quiescent potential at the gate of M_1 (M_3), reducing the headroom available for M_2/M_6 and M_4/M_5 .

The circuit of Fig. 11 robustly addresses the issues that caused problems with the scheme of Fig. 10.

• **Initial current:** The peak current is seen to be desensitized with respect to error in the voltage at the gates of M_1/M_3 due to resistive degeneration and the large $g_m R_D$ product in the initial portion of the DAC pulse. Charge sharing is less of an issue since the parasitic capacitors involved (source-bulk capacitance of the devices) are much smaller (compared to the gate-source capacitances involved in the generator of Fig. 10). The voltage drops

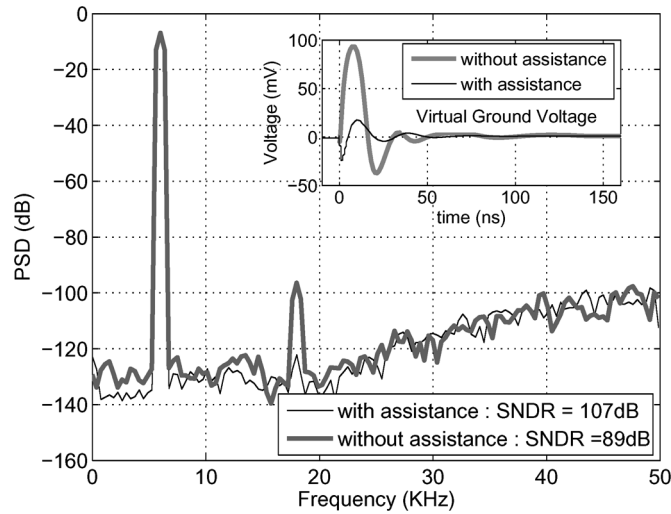


Fig. 13. Low-frequency PSD of NRZ modulators using conventional and assisted opamp integrators (with the same power consumption). The inset shows the voltages at the virtual ground node in both cases.

TABLE I
PERFORMANCE OF A CTDSM USING A CONVENTIONAL INTEGRATOR (NRZ DAC) AS A FUNCTION OF CURRENT CONSUMPTION

Total Current (μA)	Second Stage Current (μA)	In-band SNDR (dB)
27	20	89
39	32	102
47	40	103.5

across switches are no longer a problem as they are taken into account while setting the bias voltages at the gates of M_1/M_3 .

• *Time constant matching*: As explained before, the time constant of the assistant DAC pulse is matched well with that of the SCR feedback DAC waveform in the initial (high current) portion (Fig. 12). As the current reduces, the “time-constant” of the assistant pulse increases (due to the decreasing g_m of M_2/M_6 , M_4/M_5).

• *Current in the off state*: When ϕ_1 is low, the sources of M_2/M_6 and M_4/M_5 are open, ensuring zero current.

C. Simulation Results

Power Reduction using the Assisted Opamp Integrator: Simulations were run to estimate the benefit (power reduction) of using opamp assistance in the NRZ and SCR-DAC modulators. To do this, the modulator was simulated by replacing the assisted opamp integrator with a conventional active-RC integrator. The two-stage feedforward topology of Fig. 7 was used for the opamp, keeping the first stage current fixed at $5\mu\text{A}$, and scaling the second stage (by maintaining the same gate overdrives) to accommodate the current that the opamp needs to source/sink. The input to the modulator was a -5 dBFS , 6 kHz sinewave. In the NRZ case, the total power consumed by the assisted opamp integrator (opamp, assistant transconductor and DAC) is $27\mu\text{A}$, and the simulated in-band SNDR is 107 dB (this

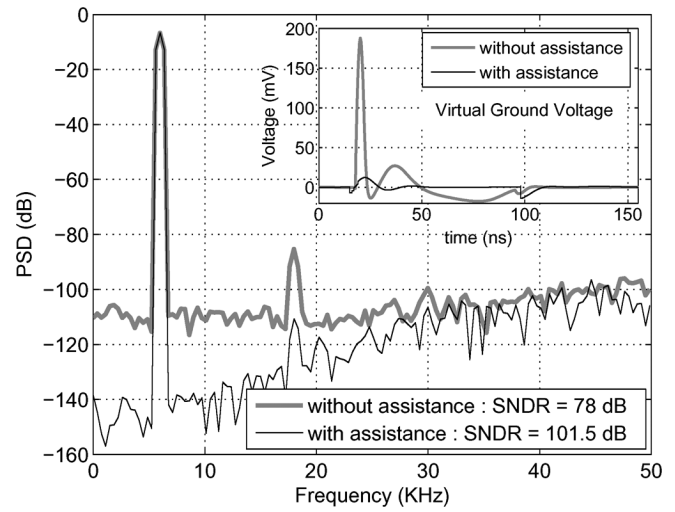


Fig. 14. Low-frequency PSD of SCR modulators using conventional and assisted opamp integrators (with the same power consumption). The inset shows the voltages at the virtual ground node in both cases.

TABLE II
PERFORMANCE OF A CTDSM USING A CONVENTIONAL INTEGRATOR (SCR DAC) AS A FUNCTION OF CURRENT CONSUMPTION

Total Current (μA)	Second Stage Current (μA)	In-band SNDR (dB)
35	28	78
67	60	98
113	106	102

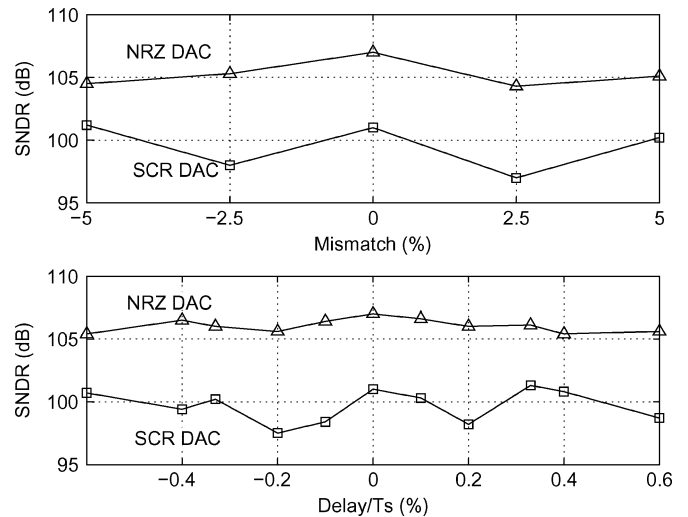


Fig. 15. Simulated performance of modulators employing assisted integrators with NRZ and SCR DACs, with $\pm 5\%$ mismatch and $\pm 1\text{ ns}$ ($0.6\% T_s$) timing skew between feedback and assistant DACs.

is almost the simulated SNDR of a modulator with a linear loop filter). When the assistant is removed, and the quiescent current in the second stage of the opamp increased to $10\mu\text{A}$ per leg (so that the current consumption is the same as the assisted integrator), the inband SNDR is 89 dB . The PSD of both modulators (with and without assistance, but same total power) is shown in

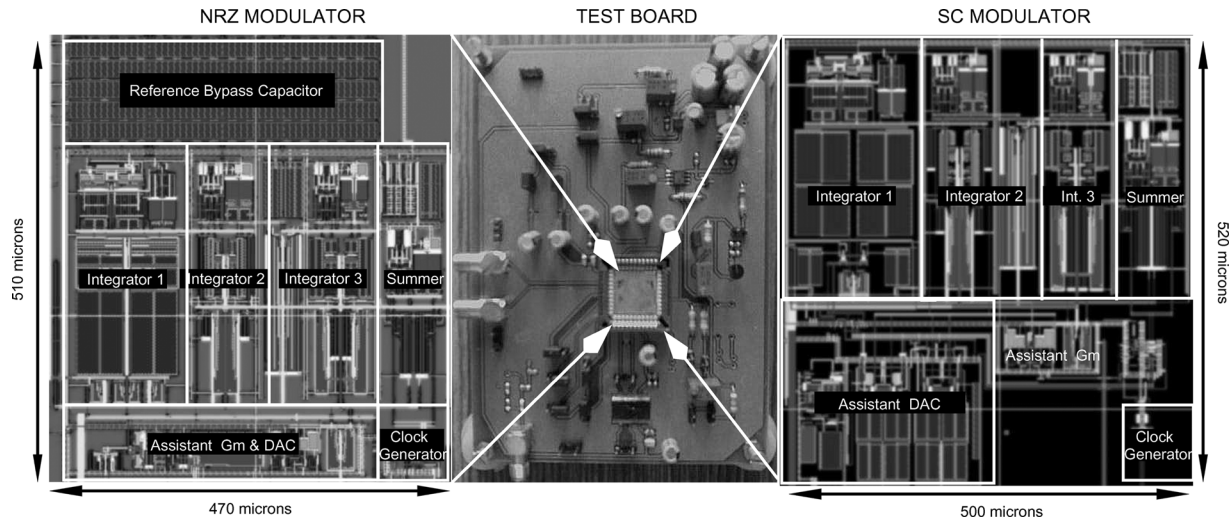


Fig. 16. Test board and chip layouts of the NRZ and SCR DAC based modulators.

Fig. 13. The inset in the figure shows the virtual ground waveforms in both cases—it is seen that the voltage excursions are considerably smaller with assistance. The second stage current was further increased (in the conventional integrator) to determine the improvement possible with enhanced power dissipation. The results are given in Table I. The second row of the table corresponds to the usual practice of designing the opamp so that slewing is avoided (with some design margin). The maximum load current that the opamp needs to supply per leg is $13.5 \mu\text{A}$. Using a design margin of $2.5 \mu\text{A}$ per leg in the second stage, a first stage current of $5 \mu\text{A}$ and CMFB/bias current of $2 \mu\text{A}$ (the same as in the assisted case), the resulting current consumption is $39 \mu\text{A}$. Still, the in-band SNDR is not quite as high as in the assisted modulator. It is thus seen that significantly more current is needed in a class-A opamp to achieve low distortion operation when compared to the assisted opamp design (which needs only $27 \mu\text{A}$).

The same experiments were run with the SCR-DAC based modulator. Fig. 14 compares the PSD of modulators with and without assistance, for the same power consumption in both cases (about $35 \mu\text{A}$). It is seen that using assistance improves the SNDR by about 23.5 dB. Table II gives the simulated performance of the SCR-DAC based modulator (without assistance) as a function of the total bias current, where the third row represents the situation where currents in the second stage of the opamp were chosen to avoid slewing.

From Tables I and II, it is seen that a conventional opamp would have to dissipate significantly higher power to approach the performance achieved using an assisted opamp. Further, by using an assisted integrator, one can bypass the “trial and error” approach usually employed during the design of the first opamp in a CTDSM. Thus, our technique provides a simple, straightforward and power efficient design methodology.

Mismatch and Timing Errors: Since the assisted opamp integrator is based on cancellation of the swing at the virtual ground of the opamp, a potential problem is the sensitivity to mismatch and timing skew errors between the currents injected by the feedback and assistant DACs. Mismatch and skew errors in the assistant current can be modelled as the sum of the ideal current and an error current injected at the opamp output. As long

as these errors are small, it is seen that they should not significantly increase the swing at the virtual ground node, and should therefore not adversely effect the modulator performance. Further, the error current is injected at the output of the opamp, so its in-band component should be suppressed by the in-band $G_m(s)$ of the opamp, in a manner similar to the suppression of noise and distortion from the assistant. Since quantitative estimation of modulator performance with errors seems difficult, simulations incorporating mismatch ($\pm 5\%$) and timing skew ($\pm 0.6\% T_s = 1 \text{ ns}$) between the feedback and feedforward DACs were run. Fig. 15 shows the results, which indicate that the assisted opamp technique is robust in the face of practical nonidealities.

V. MEASUREMENT RESULTS

The third-order continuous-time $\Delta\Sigma$ modulators were fabricated in a $0.18 \mu\text{m}$ CMOS process through Europractice. Fig. 16 shows the test board and layout screen shots of the active areas of the CTDSMs. The NRZ and SCR designs occupy about 0.24 mm^2 and 0.26 mm^2 , respectively. This is a third of the area occupied by our multibit modulator in the same process reported in [6]. Five hundred thousand samples from the modulator output were captured using an Logic Analyzer. The data was processed offline on a PC. Since the signal source and the sampling clock were not in sync, a 32 K Blackman–Harris window (rather than the Hann window used during simulations) was used to minimize leakage effects during power spectral density estimation.

Fig. 17 shows the measured SNR/SNDR of the modulators. The SNR is determined using a 15 kHz sinewave input, while a 6 kHz tone is used for SNDR measurement. The peak SNR/SNDR are 91 dB/88 dB and 90.3 dB/89.1 dB for the NRZ/SCR modulators. The measured dynamic ranges are 92.5 dB and 91.5 dB, respectively. The PSD plots of the modulator outputs are shown in Fig. 18. Part (a) of the figure shows the spectrum produced by the CTDSM with the NRZ DAC for a -2.2 dBFS , 6 kHz sinewave input (this is the amplitude that results in the best SNDR). It is seen that the harmonics are about 95 dB below the fundamental and no nonharmonic tones are observed above the noise floor. Fig. 18(b) shows the

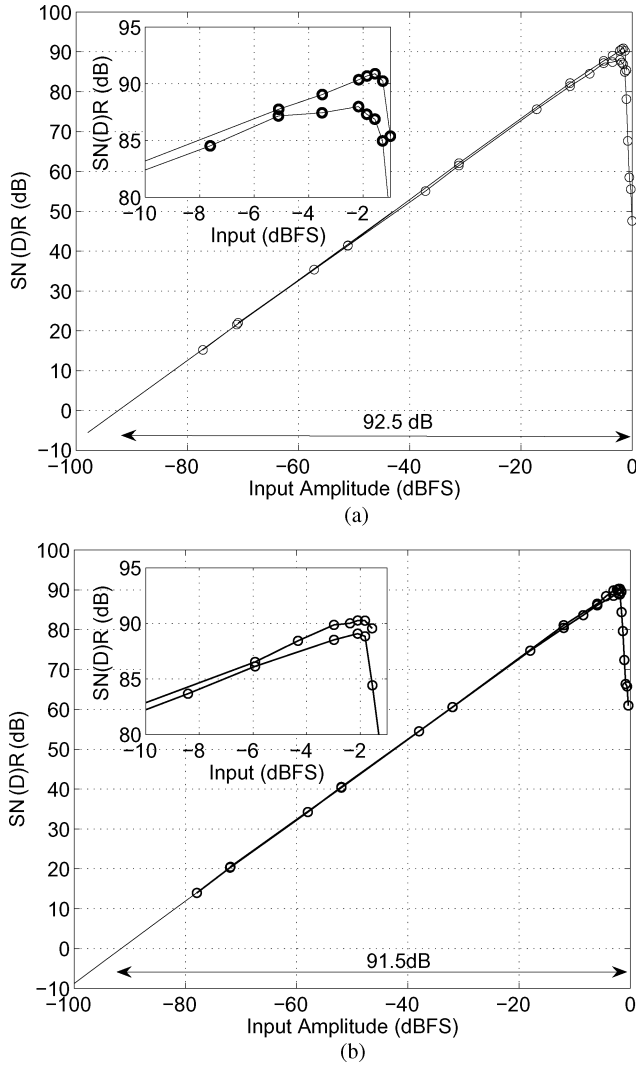


Fig. 17. Measured SNR and SNDR for the modulator with (a) an NRZ DAC and (b) an SCR DAC. The dynamic ranges are 92.5 dB and 91.5 dB, respectively.

output spectrum of the modulator with the SCR-DAC for a -2.1 dBFS 6 kHz input. The third harmonic is seen to be about 97 dB below the fundamental.

Fig. 19 shows the measured rejection in the alias-band. The simulated rejection (as computed from an AC response) is also shown for comparison, and good agreement is seen. Around f_s , the measured rejection is around 90 dB, falling to 72 dB at the edge of the alias band (where the simulated rejection is 78 dB). Fig. 20 shows the measured (and simulated) STF at out-of-band frequencies. Due to the nonlinear nature of the single-bit quantizer, the STF is a strong function of the amplitude of the sinusoidal input tone. The STF peaks significantly (15 dB) when a 50 mV input is used, but reduces to 5 dB when the amplitude is increased to 500 mV.

A. Jitter Sensitivity Measurements

Since a controlled jitter injection setup was not available, the modulators were tested using an FM modulated sinusoidal clock source, which can be expressed as

$$V_{\text{clk}} = A_{\text{clk}} \sin \left(2\pi f_s t + \frac{\Delta f}{f_m} \sin(2\pi f_m t) \right) \quad (11)$$

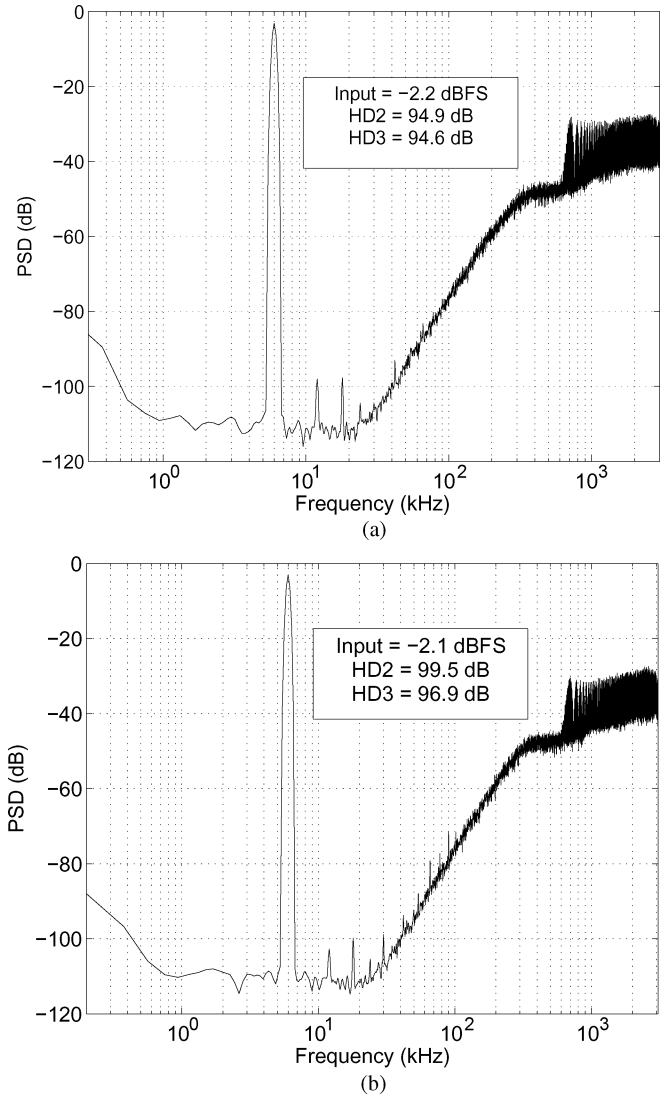


Fig. 18. Modulator output spectrum for a 6 kHz tone (a) for an NRZ-DAC based modulator with -2.2 dBFS input and (b) for an SCR-DAC based modulator with -2.1 dBFS input.

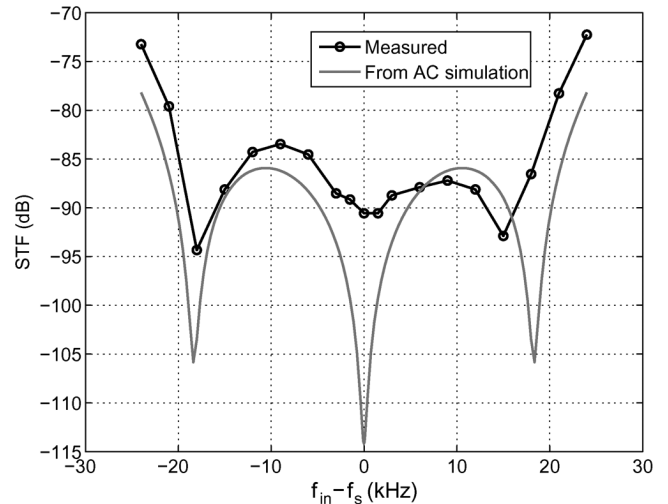


Fig. 19. Measured and simulated rejection in the first alias band ($(f_s - 24$ kHz) to $(f_s + 24$ kHz)).

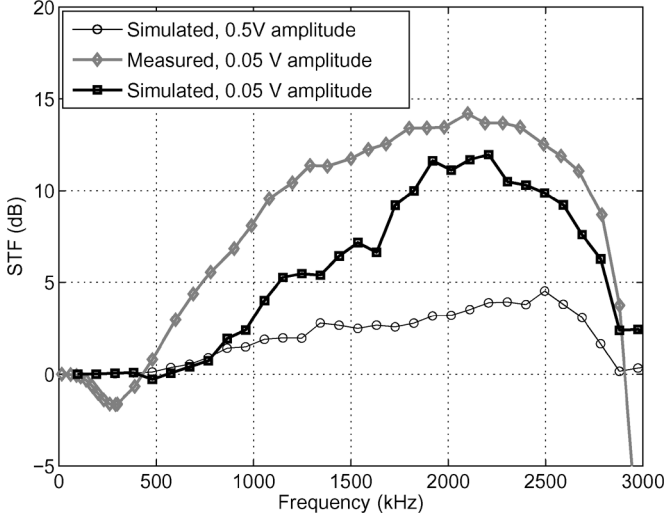


Fig. 20. Measured and simulated STF for out-of-band frequencies. The magnitude of the STF depends significantly on the input amplitude.

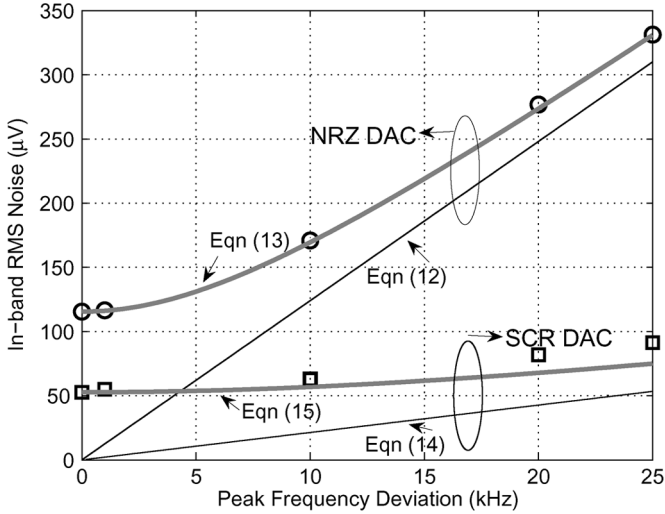


Fig. 21. Measured jitter performance of the NRZ (○) and SCR (□) modulators, with an FM modulated sinusoidal clock source. The modulating signal is a 2.5 MHz tone. The upper and lower curves in gray are calculations using (13) and (15), respectively.

where f_s and f_m refer to the sampling and modulating frequencies, and Δf denotes the peak frequency deviation. f_m was chosen to be 2.5 MHz. The in-band noise of the converters was measured as a function of Δf (with their inputs set to zero). $\Delta f = 25$ kHz corresponds to a peak-to-peak jitter of about 0.52 ns. For the modulator with the NRZ DAC, if $f_m \gg f_s/\text{OSR}$, it can be shown (by proceeding along the lines of [11]) that the RMS in-band noise due to sinusoidal jitter is white and has an RMS value given by

$$J_{\text{rms,NRZ}} = 2V_{\text{ref}} \frac{\Delta f}{\pi f_m} \sqrt{\frac{\Lambda}{8\text{OSR}}}. \quad (12)$$

In the above equation, V_{ref} is the reference voltage ($= 1.8$ V) and Λ is the probability of the quantizer making a transition. Our measurements show that $\Lambda \approx 0.75$. Fig. 21 shows the measured in-band noise as a function of Δf (denoted by the circles).

The expected jitter noise, computed from (12) is also shown for comparison. The clock source has significant jitter even when Δf was set to zero. This, and the input-referred thermal noise of the modulator, explain the measured 116 μV RMS noise for $\Delta f = 0$. This is several times higher than the noise seen with the (clean) square wave clock source used for all the measurements reported in the previous subsection. However, when $\hat{J}_{\text{rms,NRZ}}$, defined by

$$\hat{J}_{\text{rms,NRZ}} = \sqrt{J_{\text{rms,NRZ}}^2 + (116 \mu\text{V})^2}, \quad (13)$$

is plotted as a function of Δf (upper gray curve in Fig. 21), we see that there is an excellent match between prediction and measurement. This makes sense, because the thermal noise of the modulator, noise due to the intrinsic jitter of the clock source, and that due to the FM tone are uncorrelated.

With an SCR DAC, jitter effects the position of the DAC pulse, as well as its width. It is well known that pulse-position jitter is first-order noise shaped out of the signal band, and that pulse-width jitter causes the inband jitter noise to be white. The measured in-band noise with jitter is shown (denoted by squares) in Fig. 21. We see that it is significantly smaller than that in the NRZ case, indicating the superior performance of the SCR-DAC, when there is clock jitter.

The RMS in-band noise for FM jitter (with $f_m \gg f_s/\text{OSR}$) in an SCR-DAC, can be approximately shown to be

$$J_{\text{rms,SC}} = \beta V_{\text{ref}} \frac{\Delta f}{\pi f_m} \frac{I_1}{I_{\text{NRZ}}} \sqrt{\frac{1}{8\text{OSR}}} \quad (14)$$

where $\beta = |1 - \exp(j\pi f_m/f_s)|$. I_1 (simulated at 1.5 μA) denotes the current at the end of ϕ_1 (see Fig. 1) and I_{NRZ} denotes the current of an NRZ DAC that supplies the same charge as the SCR DAC in one clock cycle (in our case, it is 6 μA). When

$$\hat{J}_{\text{rms,SC}} = \sqrt{J_{\text{rms,SC}}^2 + (53 \mu\text{V})^2} \quad (15)$$

is plotted as a function of Δf , we see that there is a disagreement between predictions and measurements for values around $\Delta f = 20$ kHz. A possible reason for this is the deviation of I_1 from the value predicted in simulation, due to RC time constant variations.

A summary of measured performance is given in Table III. The figure of merit (FOM) of the converter is determined as [1]

$$\text{FOM}_{\text{SNDR}} = \frac{P}{2 \times f_B \times 2^{(\text{SNDR}-1.76)/6.02}} \quad (16)$$

where P and f_B denote the power dissipation and signal bandwidth, respectively. Since the noise performance is dominated by thermal noise, it is also relevant to compute the FOM proposed by Schreier [12] given by

$$\text{FOM}_{\text{DR}} = \text{Dynamic Range}_{\text{dB}} + 10 \log_{10} \left(\frac{BW}{P} \right).$$

Note that FOM_{DR} is a logarithmic measure. Our NRZ and SCR based converters achieve FOM_{SNDR} of 112 fJ/level and 109 fJ/level, and FOM_{DR} of 175.9 and 174.4, respectively. The FOMs are close to that achieved in the multibit design reported in [6], but with one-third the active area.

TABLE III
SUMMARY OF MEASURED PERFORMANCE OF THE NRZ AND SCR MODULATORS

	NRZ Modulator	SCR Modulator
Signal Bandwidth/Clock Rate	24 kHz / 6.144 MHz	24 kHz / 6.144 MHz
Quantizer Range	3.6 V _{pp,diff}	3.6 V _{pp,diff}
Input Swing for peak SNR	-1.6 dBFS	-2 dBFS
Dynamic Range/SNR/SNDR	92.5 dB/91 dB/88 dB	91.5 dB/90.3 dB/89.1 dB
Active Area	0.24 mm ²	0.26 mm ²
Process/Supply Voltage	0.18 μ m CMOS/1.8 V	0.18 μ m CMOS/1.8 V
Power Dissipation (Modulator + References)	110 μ W	122 μ W
Figure of Merit (SNDR)	112 fJ/level	109 fJ/level
Figure of Merit (Schreier)	175.9	174.4

TABLE IV
COMPARISON WITH OTHER $\Sigma\Delta$ MODULATORS

Reference	B/W (kHz)	OSR	Type (DT/CT)	Dynamic Range/SNR/SNDR (dB)	Power (μ W)	Tech. (μ m)	Supply (V)	FOM _{SNDR} (pJ/level)	FOM _{DR} (dB)
[13]	24	128	DT	92/91/89	1500	0.13	0.9	1.36	164
[17]	20	100	DT	88/85/81	140	0.09	1	0.38	169.5
[14]	20	50	DT	83/82.2/73.1	60	0.13	0.9	0.406	168.2
[15]	20	100	DT	85/84/81	36	0.18	0.7	0.098	172.4
[16]	25	100	DT	100/100/95	870	0.18	0.7	0.38	174.6
[1]	25	48	CT	80/73/70	135	0.5	1.5	1.04	162.7
[2]	25	48	CT	81/66/66	250	0.5	1.5	3.07	161
[4]	20	128	CT	106/106/98	18000	0.35	3.3	6.93	166.5
[5]	20	300	CT	95/77/ 74	2200	0.065	1.2	13.43	164.6
[18]	3.4	64	CT	80/-/70	210	0.5	2.2	11.95	152.1
[19]	25	64	CT	74/74/74	300	0.18	0.5	1.46	153.2
[6]	24	64	CT	93.5/92.5/90.8	121	0.18	1.8	0.089	176.5
This work (NRZ)	24	128	CT	92.5/91/88	110	0.18	1.8	0.112	175.9
This work (SCR)	24	128	CT	91.5/90.3/89.1	122	0.18	1.8	0.109	174.4

B. Comparison With Other Work

Table IV compares the performance of several $\Delta\Sigma$ modulators presented in the literature with the ADC presented in this work. We use the FOM to assess power efficiency. References [13]–[17] are discrete time designs. References [1] and [2] are single-bit multiple feedback continuous-time designs that achieve SNDRs of 70 and 65 dB, respectively. The latter uses an SCR feedback DAC. It appears that the linearity of these modulators is limited by the first integrator. The ADC presented in [4] employs a four-bit quantizer in the loop, and has a hybrid continuous-time and switched-capacitor loop filter. Apart from the modulator, the chip also includes RC time constant tuning loops and a “clock clean-up” loop that makes the performance insensitive to clock jitter. [5] presents a design in a 65 nm CMOS process. It employs a “tracking ADC” to reduce the number of comparators in the

4-bit quantizer. Due to the high OSR, the power efficiency is low. The modulator in [18] employs a single-bit modulator and is intended for voice applications. Reference [19] is a single-bit design in a 0.5 V supply, but yields 74 dB SNDR. The power consumption is high due to the low supply voltage, and the linearity is limited due to single-bit operation and the linearity of the first integrator. Reference [6] is a 15-bit multibit continuous-time design that consumes a power of 121 μ W (including reference buffers), and achieves a FOM of 65 fJ/level. It is seen that the NRZ modulator described in this work achieves a FOM which is better than that of single bit CTDSMs reported earlier, with an efficiency that approaches that of the modulator in [6], which uses a 4-bit quantizer and occupies thrice the area. The SCR-DAC based modulator is power efficient and achieves very low distortion, thanks to the assisted opamp technique.

C. Prospects and Related Work

The designs presented in this paper were CIFF modulators with a single bit quantizer. The assisted opamp technique described in this paper is fundamentally a way of improving the performance of an integrator—it should benefit other modulator topologies (like CIBF) as well. While this paper demonstrates the efficacy of the “assisted opamp” idea for NRZ and exponentially decaying pulses, the benefits carry over to other DAC pulse shapes (like RZ DACs). This is because the cancellation of the swing at the virtual ground of the opamp is independent of the integrator input waveforms (as long as there is good matching between the assistant and the net input current flowing into the integrating capacitor).

The first opamp is perhaps the most challenging block in a single-bit CTDSM design. The usual approach taken is to choose an opamp topology, and iterate the slew rate and bandwidth until the SNDR of the modulator meets specifications. The assisted opamp technique is a systematic method that addresses this challenge. The intuition behind this technique is to use feedforward in a manner as to reduce the swings at all the transistor inputs in the opamp (ideally to zero), thereby resulting in low distortion operation (while reducing power and without a noise penalty). It takes the guesswork out of the design process.

Multibit operation is another technique that greatly reduces the current that needs to be supplied by the first integrator. The concept of assistance, though applicable in such situations, might not be necessary. Mismatch and timing skew will obviously reduce the benefits of “assisted” operation at higher modulator speeds.

The idea of “helping” an opamp has been attempted before, in the context of a switched capacitor integrator [20]. In that work, the authors use a single-stage folded cascode opamp and a “booster” circuit which supplies a large portion of the current that the opamp would otherwise need to supply. This is done by sampling the integrator input on a small capacitor, converting this to a current and amplifying this using a current-mirror network before supplying it to the opamp output. While this might be suitable for a switched capacitor application (since only the settled value is of interest), the delay in the current-mirror network can significantly impair a continuous-time design. Reference [20] gives experimental results showing enhanced slew rate in an integrator, but does not apply it to more functional blocks like filters or DSMs.

VI. CONCLUSION

The power dissipation in the first opamp of a conventional single bit continuous-time oversampled converter is large due to high bandwidth and linearity requirements. This becomes an even bigger problem when a SCR feedback DAC is used. The assisted opamp integrator, which is a circuit technique that addresses this difficulty, was introduced in this paper. This approach enables low distortion operation without affecting noise performance, with a significant *reduction* in the opamp’s bias current. This technique was applied to the design of audio-frequency third-order CTDSMs (with NRZ and SCR feedback DACs) implemented in a 0.18 μm CMOS technology. The NRZ/SCR ADCs have a dynamic range of 92.5 dB/91.5 dB while dissipating 110 μW /122 μW from a 1.8 V supply, with a power efficiency comparable to the best multibit modulators reported with similar specifications.

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